

WHAT AUTONOMOUS DRIVING Really Means For India

India's autonomous future will not arrive overnight, and it will not look like anyone else's. But step by step, through semi-autonomy, smart policy, and local innovation, AI can make Indian roads safer while still keeping humans firmly in the driver's seat.



An AI-powered autonomous vehicle on an Indian city road with mixed traffic (Image created by the author using OpenAI DALL·E)



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Artificial intelligence (AI) has evolved from its early role in mobile robot navigation to the advanced technologies that underpin fully autonomous vehicles. In India, this evolution extends beyond technological curiosity to directly address road safety, traffic congestion, and energy sustainability within one of the world's most complex transport ecosystems.

From mobile robots to driverless cars: The evolution of AI

The story of AI's journey from enabling mobile robots to powering autonomous vehicles is one of relentless innovation. Early mobile robots relied on basic sensors and algorithms to navigate controlled environments such as factories and warehouses. These systems followed predefined paths while avoiding obstacles. However, their limited ability to adapt to dynamic settings exposed the need for more advanced AI-driven approaches.

As deep learning and computer vision matured, AI has enabled machines to

operate in unstructured environments. These developments laid the foundation for autonomous vehicles, where AI systems function at significantly higher levels of complexity. Modern driverless cars use convolutional neural networks (CNNs) and advanced machine learning models to identify objects, predict behaviour, and make split-second decisions. In parallel, advances in sensor fusion have improved how autonomous systems integrate inputs from lidar, cameras, and radar, enabling them to build a more comprehensive understanding of their surroundings.

India's road conditions pose distinct challenges for these technologies. Road environments vary widely, from well-paved highways to congested urban streets shared by pedestrians, cyclists, and stray animals. Autonomous systems must therefore cope with highly unstructured and unpredictable conditions. In addition, seasonal weather patterns, particularly monsoons that reduce visibility and lead to waterlogged roads, further complicate autonomous navigation across the country.